MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
PROCTORED ONLINE MAKEUP EXAMINATIONS, FEBRAUARY 2022

REAL TIME SYSTEMS [ELE4064]
REVISED CREDIT SYSTEM

Time: 75 Minutes + 10 Minutes **Date: 24 February 2022** **Max. Marks: 20**

Instructions to Candidates:
- Answer **ALL** the questions.
- Missing data may be suitably assumed.
- Time: 75 minutes for writing + 10 minutes for uploading.

1A. Obtain the task schedule for the following independent periodic tasks till t = 60 using

i. Rate monotonic algorithm

ii. Deadline monotonic algorithm

T_1= (5,1.5), T_2= (10,2,10), T_3= (25,5,30), T_4= (20,4,40), and T_5= (50,10,20).

Analyze the schedule obtained and determine the feasibility of the schedule. Justify your answer. ***(04)***

1B. Schedule the following tasks using RMA **i.** without slack stealing **ii.** with slack stealing.

Periodic tasks: T_1 = (3,1.1), T_2 = (10,3.9); Aperiodic task: 'A' released at t = 0.2 and execution time of 1.8.

Which of these methods is better for scheduling aperiodic jobs? Justify your answer. ***(03)***

1C. Obtain a feasible schedule for the real time task sets (T1, T2, …T5) with precedence shown in Fig.Q1C, using a table driven scheduler. T1 = (0,9,3,9), T2 = (0,25,6,25), T3 = (0,24,7,24), T4 = (0,50,11,50), and T5 = (0,46,8,46). Describe the method used.

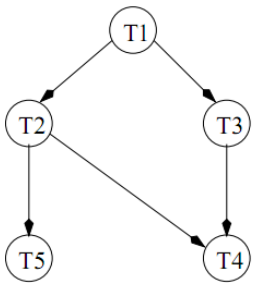


Fig. Q.1C ***(03)***

2A. With the help of a relevant pseudocode, describe the priority inheritance protocol. Discuss the limitations of priority inheritance protocol. ***(03)***

2B. Use analytical method of time demand analysis to check whether the following periodic tasks can be feasibly scheduled under RMA on a uniprocessor. T_1 = (3,1.2), T_2 = (4,1.4), and T_3 = (6,0.6). Verify and validate your answer using graphical method of time demand analysis. ***(04)***

2C. A network designed using IEEE802.4 protocol has three nodes, N1, N2, and N3:

N1 needs to transmit 1.2 MB of data every 225 msec.

N2 needs to transmit 0.8 MB of data every 400 msec.

N3 needs to transmit 2.5 MB of data every 250 msec.

Ignore propagation time and

i. Select suitable TTRT. Justify your answer.

ii. Determine the synchronous bandwidth of the nodes or token holding time. ***(03)***